Module 5: NATURAL LANGUAGE PROCESSING & GEN AI

Course Code: 22DS102006 - Text Analytics, Attention Transformers & LLMs

1. Text Processing & Word Embeddings

Tokenization, byte-pair encoding (BPE), TF-IDF vectorization, Word2Vec skip-gram, GloVe distributed semantic spaces, and cosine similarity retrieval.

2. The Transformer Architecture

Scaled dot-product attention, multi-head self-attention mechanisms, positional encodings, layer normalization, and feedforward networks. BERT encoder representations and GPT autoregressive decoders.

3. Modern Generative AI & Vector Search

Foundation models, prompt engineering, few-shot prompting, Retrieval-Augmented Generation (RAG) using dense vector databases (Chroma, Pinecone, FAISS), and fine-tuning with LoRA.